

The mixed dominating set problem is MAX SNP-hard

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Abstract

Given a graph $G = (V, E)$, a *mixed dominating set* MD of G is defined to be a subset of $V \cup E$ such that every element in $\{(V \cup E) \setminus MD\}$ is either adjacent or incident to an element of MD . The *mixed dominating set problem* is to find a mixed dominating set with minimum cardinality. This problem is NP-hard. In this paper, we prove that this problem is MAX SNP-hard.

Keywords: MAX SNP-hard, NP-hard, approximability class, vertex dominating set, edge dominating set, mixed dominating set

1 Introduction

The (*vertex*) *dominating set problem* is one of the fundamental problems in graph theory [3, 8, 10, 11, 14]. Given an arbitrary graph $G = (V, E)$ with vertex set V and edge set E , an edge $e \in E$ (respectively, a vertex $v \in V$) is said to be *incident* to a vertex $v \in V$ (respectively, an edge $e \in E$) if v is an endpoint of e . Then an edge $e \in E$ (respectively, a vertex $u \in V$) is said to be *adjacent* to an edge $e' \in E$ (respectively, a vertex $v \in V$) if edges e and e' (respectively, vertices u and v) are incident to the same vertex (respectively, edge). A *vertex dominating set* VD (respectively, an *edge dominating set* ED) of G is defined to be a subset of V (respectively, E) such that every element in $\{V \setminus VD\}$ (respectively, $\{E \setminus ED\}$) is adjacent to an element of VD (respectively, ED). The *vertex dominating set problem* (VDSP) (respectively, *edge dominating set problem* (EDSP)) is to find a vertex dominating set (respectively, edge dominating set) with minimum cardinality [3, 8, 10, 11, 14]. Both problems were shown to be NP-complete [7]. It had been shown that both problems have many important applications in chess, facility location, social network, wireless and telephone switching network routing, and so on [4, 6, 10, 13, 18, 19].

Haynes and Hedetniemi [10] surveyed some of variants of the VDSP. One of these variants is the *mixed dominating set problem* (also called the *total cover problem* in [1]). A *mixed dominating set* MD of G is defined to be a subset of $V \cup E$ such that every element in $\{(V \cup E) \setminus MD\}$ is adjacent or incident to an element of MD [10]. The *mixed dominating set problem* (MDSP) is to find a mixed dominating set with minimum cardinality [1, 10]. Zhao, Kang and Sohn [20] provided a possible application of the MDSP that is the phase measurement units (PMUs) location. They also proved that the MDSP is NP-hard, even for split graphs [20]. Given a graph $G = (V, E)$, let a total graph G_T of a graph G be a graph such that the vertex set of G_T corresponds one-to-one with the elements of $V \cup E$ and two vertices of G_T are adjacent in G_T iff their corresponding elements are either adjacent or incident in G . It is clear from the above definition that the optimal solution of G for the MDSP is equivalent to the optimal solution of G_T for the VDSP [11, 12]. However, the VDSP admits no $(c \cdot \ln |V|)$ -approximation algorithm for any $c > 0$ unless $NP \subseteq DTIME(|V|^{\log \log |V|})$ [5]. Hence, the lower bound on the best possible approximation ratio for the MDSP is $(c \cdot \ln |V|)$ for any $c > 0$ at present. However, Hatami [9] gave a 2-approximation algorithm for the MDSP in $O(|V|^3)$ time. Hence, it is still unclear whether the MDSP is MAX SNP-hard, which implies that it cannot be approximated within a ratio of $1 + \epsilon$ (polynomial time approximation scheme) for some $\epsilon > 0$ unless $P = NP$ [2]. In this paper, we prove that the MDSP is MAX SNP-hard by the *L-reduction* from the EDSP. The EDSP is known to be MAX SNP-hard [17].

2 The mixed dominating set problem is MAX SNP-hard

In this section, a graph $G = (V, E)$ is simple, connected and undirected. Let (u, v) denote an edge connecting two vertices u and v . For an edge (u, v) of G , vertices u and v are called *incident* vertices of (u, v) , and (u, v) is called an incident edge of u and v . For a vertex v of G , a vertex u is called an *adjacent* vertex of v if u and v share a common incident edge (u, v) . Then for an edge (u, v) of G , an edge (u', v') is called an adjacent edge of (u, v) if (u, v) and (u', v') share a common incident vertex. For an element $x \in \{V \cup E\}$, we let $ADJ(x)$ denote a set of all adjacent vertices or edges of x , and let $INC(x)$ denote a set of all incident vertices or edges of x . Then we will show that the MDSP is MAX SNP-hard by an *L-reduction* from the EDSP. Hence, we review the definition of the *L-reduction* as follows.

Definition 1: [15] Given two optimization problems π_1 and π_2 , we say that π_1 *L-reduces* to π_2 if there exist polynomial time computable functions f and g and positive constants α and β such that for every instance I_1 of π_1 , the following properties hold:

1. function f produces an instance $I_2 = f(I_1)$ of π_2 such that $OPT(I_2) \leq \alpha \cdot OPT(I_1)$, where $OPT(I_1)$ and $OPT(I_2)$ are the costs of optimal solutions of I_1 and I_2 , respectively.
2. Given any solution of I_2 with cost c_2 , function g produces a solution of I_1 with cost c_1 such that $|c_1 - OPT(I_1)| \leq \beta \cdot |c_2 - OPT(I_2)|$.

Now, let f be the polynomial-time function to transform an instance I_1 of the EDSP to the instance I_2 of the MDSP (i.e., $f(I_1) = I_2$). Let a graph $G = (V, E)$ be an instance I_1 of the EDSP with $V = \{v_1, v_2, \dots, v_n\}$. We transform I_1 into an instance I_2 of the MDSP, say $G_2 = (V_2, E_2)$, as follows.

$V_2 = V \cup \{v_a, v_b\}$, where v_a and v_b are two new vertices.

$E_2 = E \cup (v_a, v_b) \cup \{(v_a, v_i) | 1 \leq i \leq n\}$.

Next, we describe another polynomial-time function g to transform any solution of G_2 for the MDSP to a solution of G for the EDSP. Given a mixed dominating set MD of G_2 , we transform it into an edge dominating set ED of G by the following method.

Function g

Let a set $MD_1 = MD$.

For each $x \in MD_1$ **do**

1. **If** $x \in \{v_a, v_b, (v_a, v_b)\}$ **then** $MD_1 = \{MD_1 \setminus \{x\}\}$.
2. **If** $x \in \{(v_a, v_i) | 1 \leq i \leq n\}$ **then** $MD_1 = \{MD_1 \setminus \{x\}\}$ and add an edge $e \in \{ADJ(x) \cap E\}$ into MD_1 .
3. **If** $x \in V$ **then** $\{MD_1 = MD_1 \setminus \{x\}\}$ and add an edge $e \in \{INC(x) \cap E\}$ into MD_1 .

end for

Let $ED = MD_1$.

Lemma 1: For any mixed dominating set MD of G_2 , function g transforms it into an edge dominating set ED of G and $|ED| \leq |MD| - 1$.

Proof: Clearly, a mixed dominating set MD of G_2 must contain at least one element in $\{v_a, v_b, (v_a, v_b)\}$, and however these elements are not contained in any feasible solution of G for the EDSP. Hence, we delete it by step 1. For a mixed dominating set MD of G_2 , each edge in E is adjacent or incident to an element of MD (i.e., $\bigcup_{x \in MD} \{(ADJ(x) \cup INC(x)) \cap E\} = E$). For every element $x \in \{(v_a, v_i) | 1 \leq i \leq n\}$, we add an edge $e \in \{ADJ(x) \cap E\}$ to ED in step 2, and then we have $\{ADJ(e) \cap E\} \supseteq \{ADJ(x) \cap E\}$. Finally, for every element $x \in V$, we add an edge $e \in \{INC(x) \cap E\}$ to ED in step 3, and then we have $\{ADJ(e) \cap E\} \supseteq \{INC(x) \cap E\}$. Hence, after running function g , all elements in ED are edges and each edge in E is adjacent to an edge of ED . Hence, ED is an edge dominating set of G . Since MD must contain at least one element in $\{v_a, v_b, (v_a, v_b)\}$, we have $|ED| \leq |MD| - 1$. $\square$

Theorem 2: The mixed dominating set problem is MAX SNP-hard

Proof: We prove that functions f and g constitute an *L-reduction* from the EDSP to the MDSP by showing the following two inequalities.

(1) $OPT(I_2) \leq \alpha \cdot OPT(I_1)$, where $\alpha = 2$. Since the vertex v_a can be adjacent to all vertices in V_2 and incident to all edges in $\{(v_a, v_i) | 1 \leq i \leq n\}$ and (v_a, v_b) , the union of the optimal solution of G for the EDSP and the vertex v_a is a feasible solution of G_2 for the MDSP. Hence, we have

$OPT(I_2) \leq OPT(I_1) + 1$. Since $1 \leq OPT(I_1)$, we have $OPT(I_2) \leq 2 \cdot OPT(I_1)$.

(2) $|c_1 - OPT(I_1)| \leq \beta \cdot |c_2 - OPT(I_2)|$, where $\beta = 1$. Note that MD denotes any mixed dominating set of G_2 and ED denotes an edge dominating set of G by the output of the function g . By definition 1, $c_1 = |ED|$ and $c_2 = |MD|$, respectively. By Lemma 1, we have

$$\begin{aligned} c_1 - OPT(I_1) &\leq c_2 - 1 - OPT(I_1) \\ &\leq c_2 - 1 - (OPT(I_2) - 1) \\ &= c_2 - OPT(I_2), \end{aligned}$$

since $OPT(I_2) - 1 \leq OPT(I_1)$. $\square$

3 Conclusions

In this paper, we have shown that the MDSP is MAX SNP-hard. It would be interesting to find an approximation algorithm or the approximability class for the weighted mixed dominating set problem (i.e., a weight function on the edges and vertices of the input graph).

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